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$$(0,3 \times \text{ЭКЗ} + 0,3 \times \text{БАЗ} + \\ + 0,25 \times \text{МАЗ} + 0,15 \text{ПР})$$

$$\text{ОКР} \left(\left(0,3 \times \text{БАЗ} + \right. \right. \\ \left. \left. + 0,25 \times \text{МАЗ} + 0,15 \text{ПР} \right) / 0,7 \right) \geq 8$$

Plan

- Backpropagation
 - Fully-connected nets
 - Activation
-

$$a(x) = w_1 x_1 + \dots + w_n x_n$$

↑ ↑
weight input

$$f(x, w) \quad x \in \mathbb{R}^n \\ w \in W$$

$$\{(x_i, y_i)\}_{i=1}^l - \text{training set}$$

$L(\hat{y}, y)$ - loss func

$$\alpha(w) = \frac{1}{l} \sum L(f(x_i, w), y_i)$$

$\rightarrow \min_w$

f, L - differentiable

FCN (fully-connected Network)

MLP (Multi-layer Perceptron)

$$a(x) = \langle w, x \rangle = w^T x \quad w \in \mathbb{R}^d$$

$$z_1 = W_1 x + b_1 \quad W_1 \in \mathbb{R}^{d_1 \times d} \quad b_1 \in \mathbb{R}^{d_1}$$

$$z_2 = W_2 z_1 + b_2 \quad \dots$$

...

$$z_n = W_n z_{n-1} + b_n \quad \dots$$

$$z_n = \underbrace{W_n \dots W_2 W_1}_W x + b$$

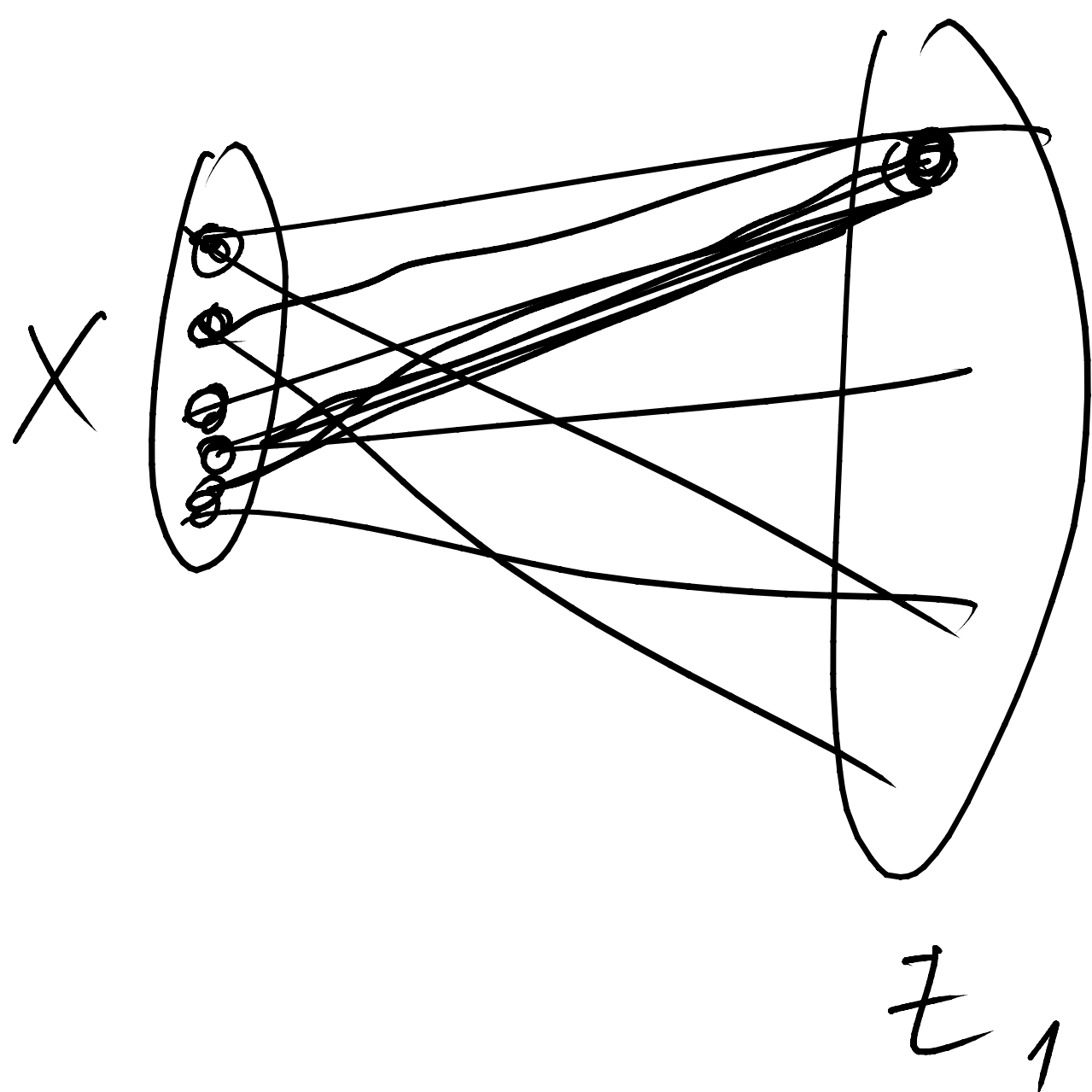
$W \in \mathbb{R}^{d_n \times d}$

$$\sigma(\cdot) : \mathbb{R} \rightarrow \mathbb{R} \quad \text{activation func}$$

- non-linear
- differentiable

$$z_i = \sigma(W_i z_{i-1} + b_i)$$

dense / linear / fc layer



Backpropagation

- Chain rule

$$1. \quad \frac{d(f(g(x)))}{dx} = \frac{df}{dg} \cdot \frac{dg}{dx}$$

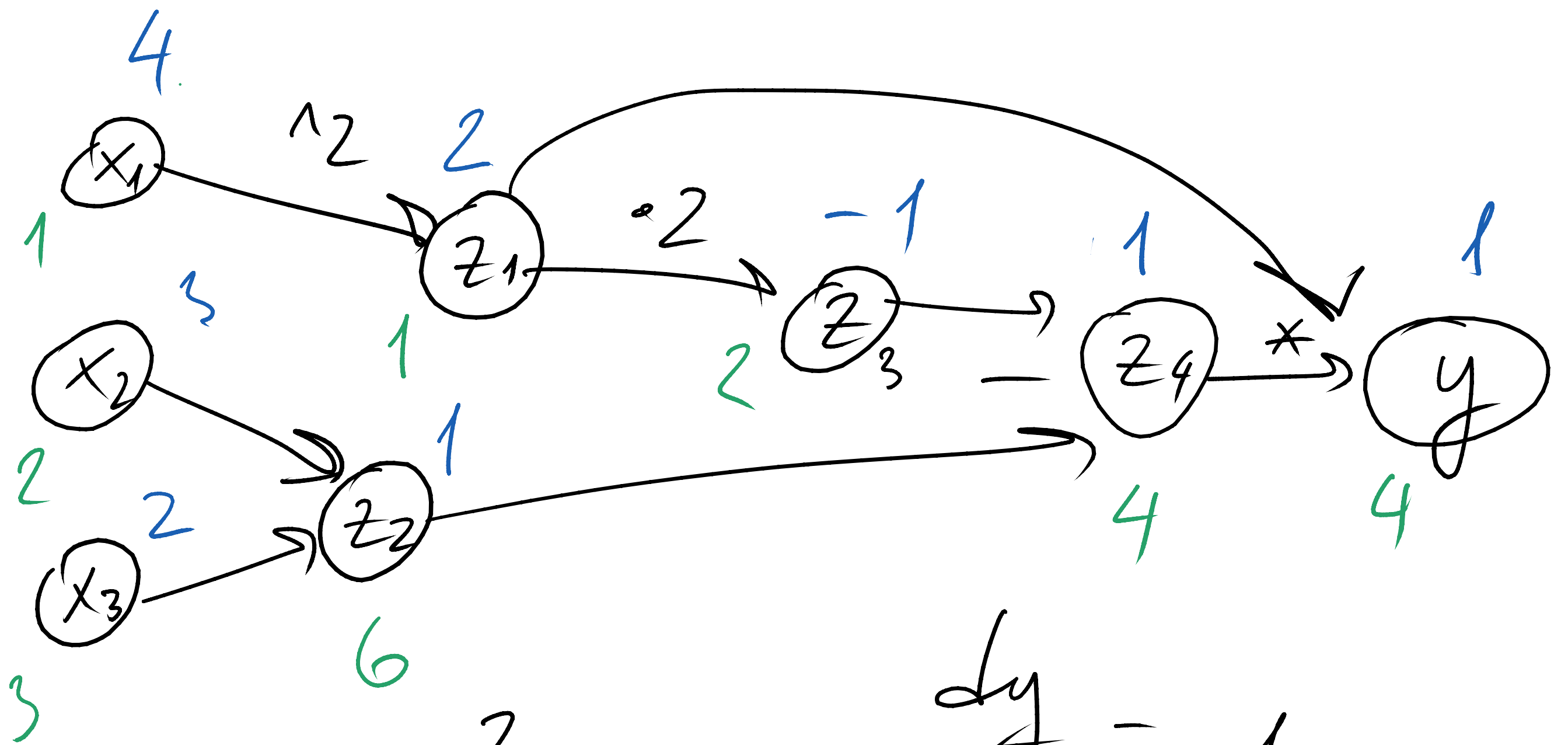
$$2. \quad f(g_1(x), g_2(x), \dots, g_n(x))$$

$$\frac{df}{dx} = \left\{ \right\} = \sum_{i=1}^n \underbrace{\frac{df}{dg_i}}_{D_i f} \cdot \frac{dg_i}{dx}$$

analytical	numerical
$(x^2)' = 2x$	$(x^2)' \Big _{x_0} = 2x_0$

$$f(x_1, x_2, x_3) = x_1^2 (x_2 x_3 - 2x_1^2)$$

$\uparrow \quad \uparrow \quad \uparrow$
 $1 \quad 2 \quad 3$



$$z_1 := x_1^2$$

$$\frac{dy}{dy} = 1$$

$$z_2 := x_2 x_3$$

$$z_3 := 2z_1$$

$$z_4 := z_2 - z_3$$

$$y := z_1 z_4$$

$$\frac{dy}{dz_4} = \frac{dy}{dy} \cdot \frac{dy}{dz_4} =$$

$$= 1 \cdot z_1 = 1 \cdot 1 = 1$$

$$\frac{dy}{dz_3} = \frac{dy}{dz_4} \cdot \frac{dz_4}{dz_3} = 1 \cdot (-1) = -1$$

$$\frac{dy}{dz_2} = \frac{dy}{dz_4} \cdot \frac{dz_4}{dz_2} = 1 \cdot 1 = 1$$

$$\begin{aligned} \frac{dy}{dz_1} &= z_4 + z_1 \frac{dz_4}{dz_1} = 4 + \frac{dz_4}{dz_3} \cdot \frac{dz_3}{dz_1} = \\ &= 4 + (-1) \cdot 2 = 2 \end{aligned}$$

$$\frac{dy}{dx_1} = \frac{dy}{dz_1} \cdot \frac{dz_1}{dx_1} = 2 \cdot 2x_1 = 4$$

$$\frac{dy}{dx_2} = \frac{dy}{dz_2} \cdot \frac{dz_2}{dx_2} = 1 \cdot x_3 = 3$$

$$\frac{dy}{dx_3} = \frac{dy}{dz_2} \cdot \frac{dz_2}{dx_3} = 1 \cdot x_2 = 2$$

$$z_1 = f_1(x, w_1)$$

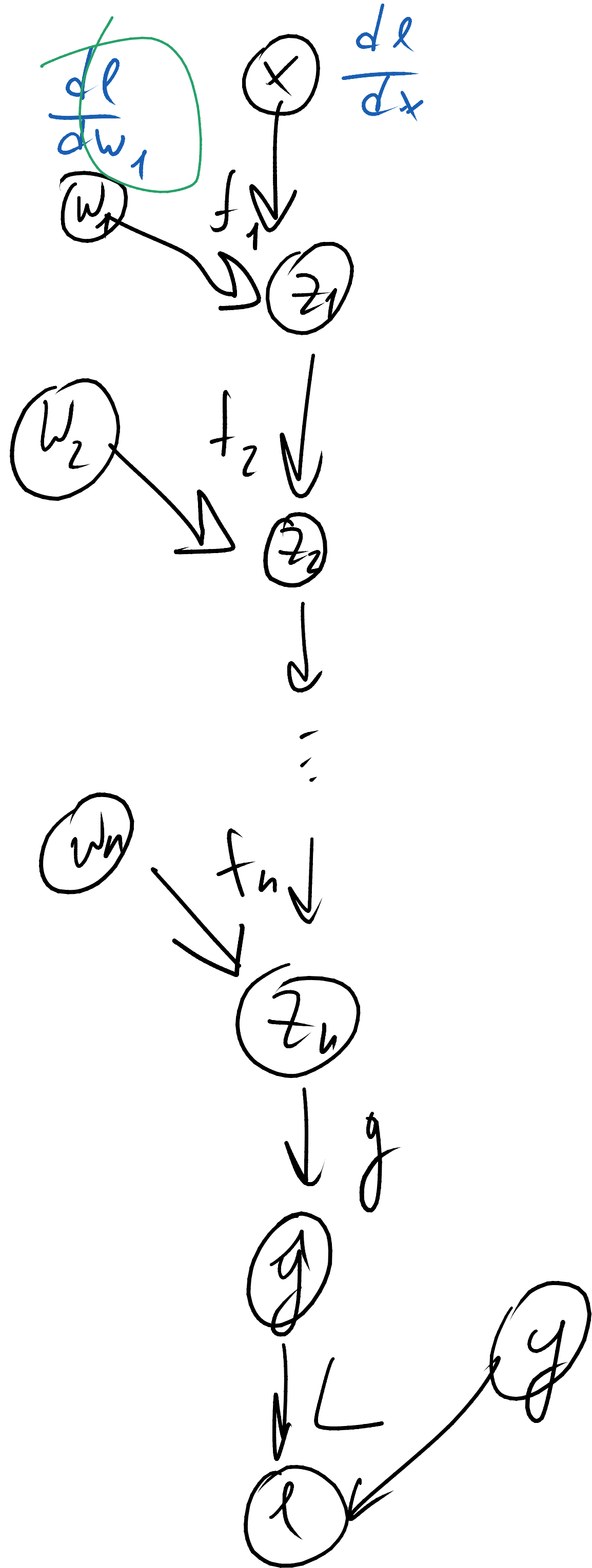
$$z_2 = f_2(z_1, w_2)$$

...

$$z_n = f_n(z_{n-1}, w_n)$$

$$\hat{y} = g(z_n)$$

$$l = L(\hat{y}, y)$$



Historic - Evolutionary alg. 1950s

- Reservoir computing (2000)
linear head

Modern — speed up, less resources

- Forward-forward (2022) post-hoc
- Direct feedback alignment (2016)

$$\left. \frac{d(f(g(x)))}{dx} \right|_{x_0} = \left. \frac{df}{dg} \right|_{g(x_0)} \cdot \left. \frac{dg}{dx} \right|_{x_0}$$

normal backprop

$$\left. \frac{d(f(g(w)))}{dw} \right|_{w_0} = \left. \frac{df}{dg} \right|_{g(w^*)} \cdot \left. \frac{dg}{dw} \right|_{w_0}$$

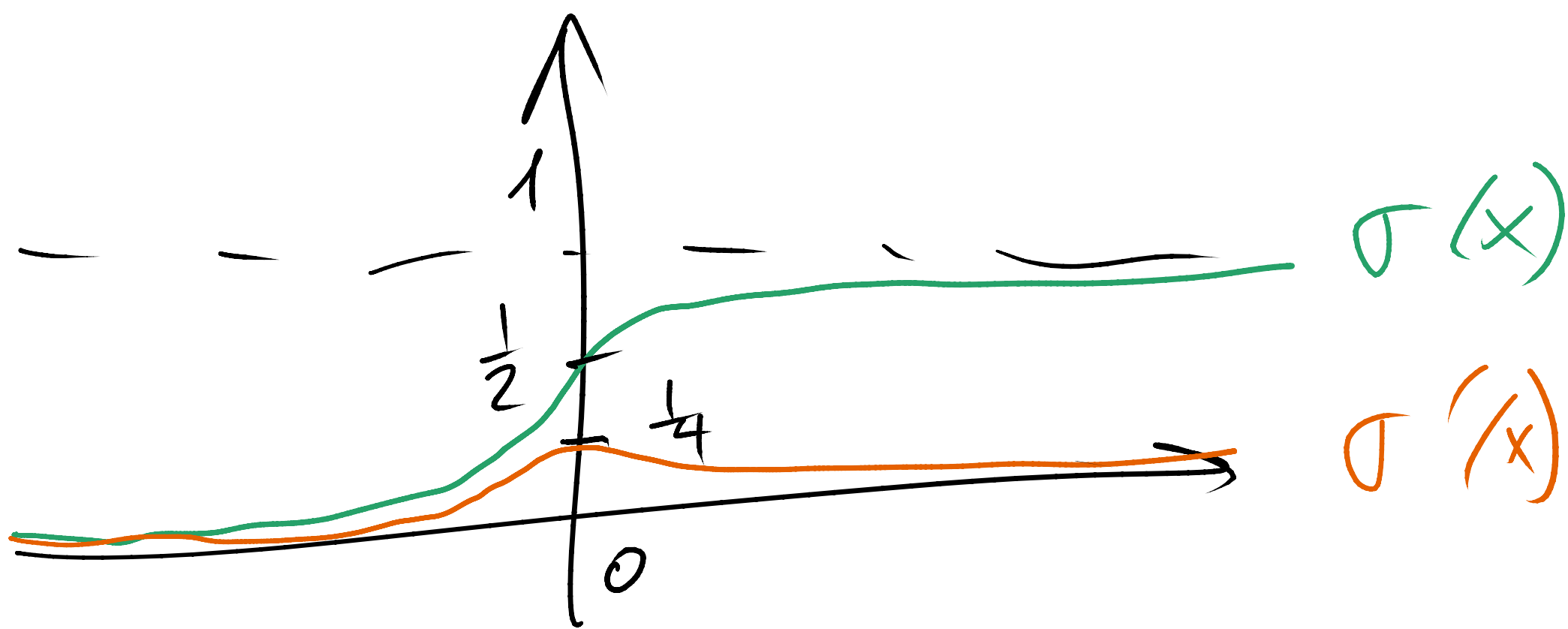
dta

w^* random
fixed
matrix (vec)

1. Sigmoid (tanh)

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\sigma'(x) = \sigma(1 - \sigma)$$

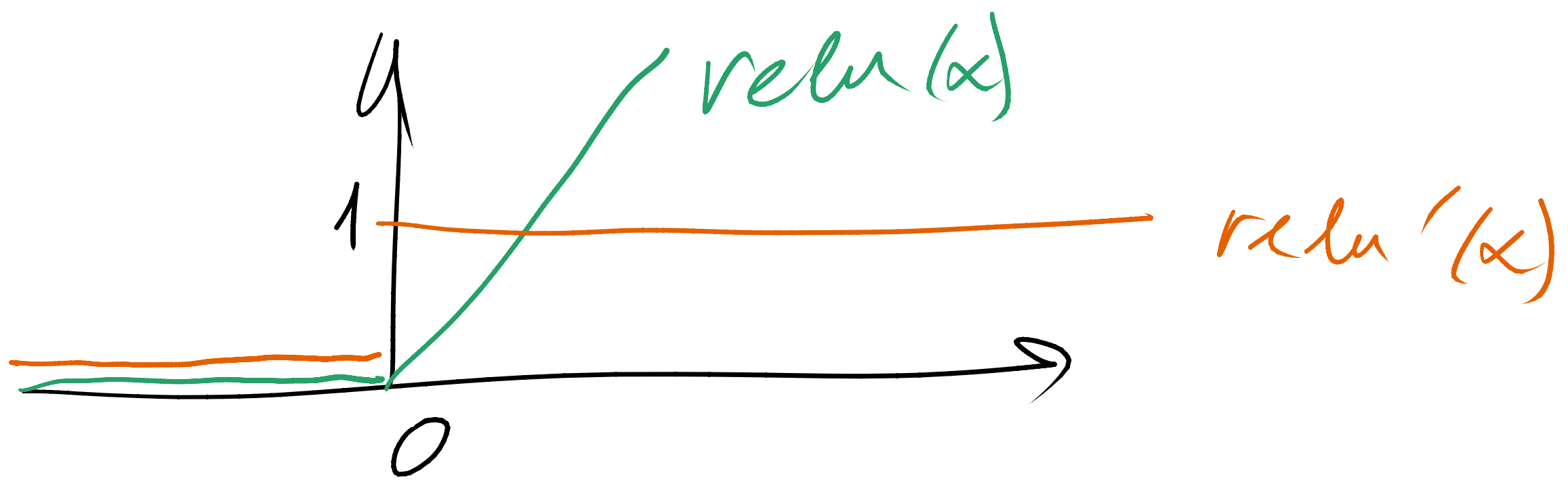


$$u_i = W_i z_i + b_i$$

$$z_{i+1} = \sigma(u_i) \quad \frac{dl}{dz_i} = \frac{dl}{dz_{i+1}} \cdot \underbrace{\sigma'(u_i)}_{\approx 0} \cdot \frac{du_i}{dz_i}$$

2. ReLU (rectified linear unit)

$$\text{ReLU}(x) = \max(0, x)$$



3. $\text{Leaky ReLU}_\alpha(x) = \max(\alpha x, x)$

